MIL-N-37175
3 February 1975

MILITARY SPECIFICATION

NEEDLE, BIOPSY, KIDNEY, MENGHINI, 6-1/8 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a biopsy needle of the Menghini design suitable for obtaining a specimen from the kidney.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.

Military

MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications and standards, required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z70.3	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New Y ork, N.Y. 10018.)

3. REQUIREMENTS

3.1 General requirements. The biopsy needle shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material.

3.2.1 Cannula. The cannula shall be fabricated from seamless drawn, or welded and drawn tubing. Shall be Class 3, type 302, 303 or 316 material.

3.2.2. Needle hub and stylet knob. The needle hub and stylet knob shall be fabricated from Class 8, type 260, 270 or 360 brass. Cavity of hub shall be nickel plated only. As an alternate shall be Class 3, type 302, 303, or 316 material.

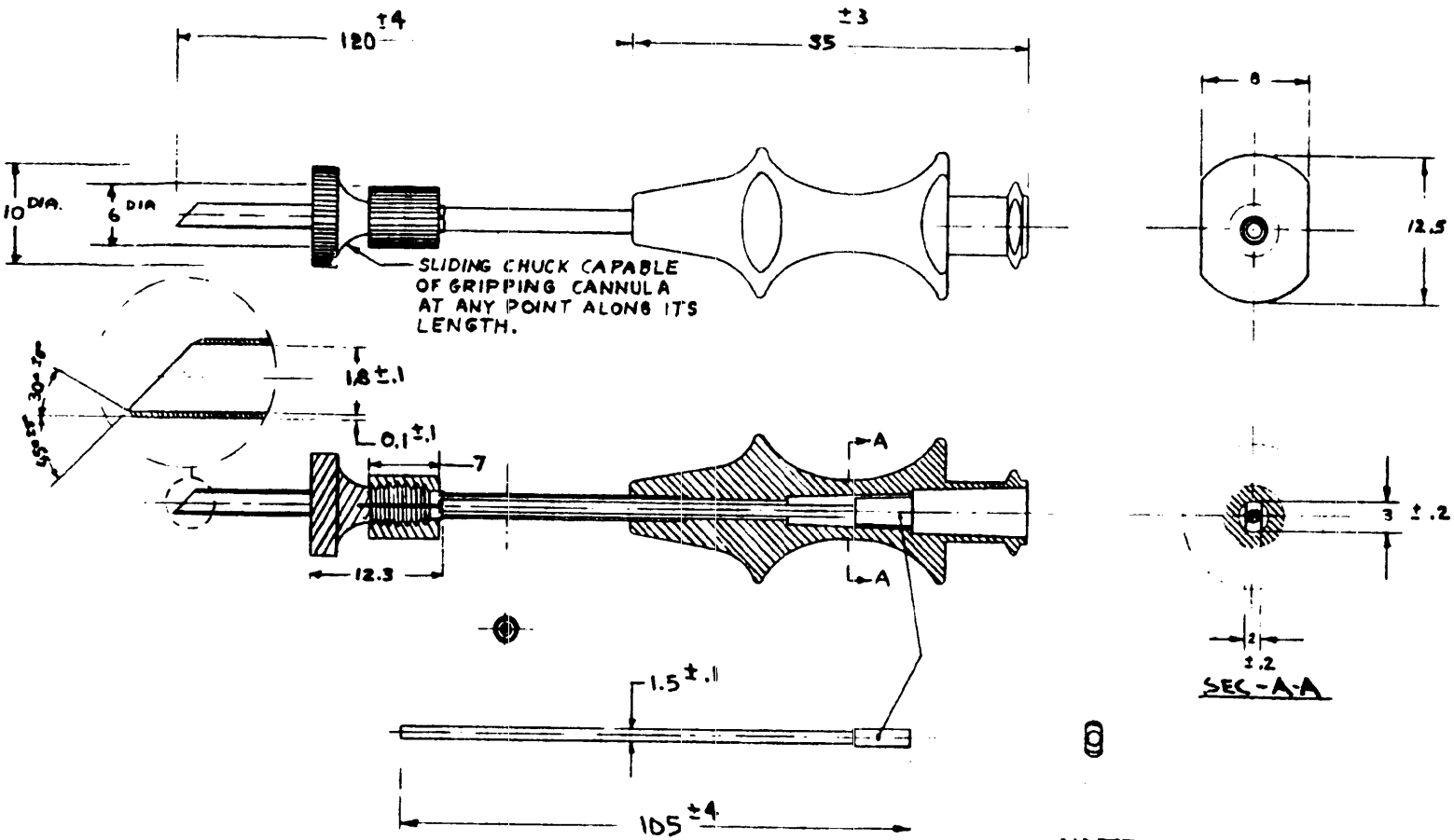
3.2.3 Stylet wire. The stylet wire shall be fabricated from Class 3, type 302, 303 or 304 material.

3.2.4 Hardness. After proper stress relieving the Rockwell hardness of Class 3 material shall be B75 to B85.

3.3 Style, design and dimensions. Shall be a biopsy needle of the Menghini design. Shall be in accordance with Figure 1.

3.4 Security of assembly. Cannula shall not detach when tested as specified in 4.4.1.

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NOTE:

1. ALL DIMENSION IN MILLIMETERS
2. UNLESS OTHERWISE SPECIFIED TOLERANCE SHALL BE ±1 MM

MIL-N-3475

NEEDLE, BIOPSY, KIDNEY, MENGHINI

FIGURE 1

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3.5 Hub. Hub shall be Luer-lock type, so shaped as to provide means for facilitating attachment to, and removal from, a syringe. Hub and hub cavity shall be in accordance with Figure 3 of ANSI Standard Z70.1. The hubs shall be tested for diameter and depth of hub cavity as specified in 4.4.2.1. Hubs which do not fit the plug within the "GO" - "NO GO" limits specified in 4.4.2 shall be considered not acceptable. In addition when the hub is fitted into the plug gage as specified in 4.4.2, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions.

3.5.1 Freedom from foreign matter. The hub cavity shall be tested for freedom from foreign matter, as specified in 4.4.2.2 and shall show complete freedom from metal chips and other foreign matter.

3.5.2 Fluid tightness. The hub shall be tested for fluid tightness, as specified in 4.4.2.3 and the soldered connections of the cannula proper with the hub and the Luer connection of the hub and the Luer connection of the hub to the plug gage shall not show leakage exceeding the formation of a drop.

3.6 Needle cannula. The cannula shall show a smooth finished outer surface free from pits and tool marks, and inner surface free from scale, deep pits, deep furrows, or die marks; the lumen of the cannula and the hub cavity shall be free from chips, dirt, or other foreign matter. Needle cannula shall be sharp not blunt or bent, and points shall be free of feathered edges, burrs, or other defects, when examined under 7X power magnification.

3.7 Finish. Finish shall be mirror or mirror and satin.

3.8 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as othersise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant metals, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one biopsy needle.

Table I. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for test purposes shall be one biopsy needle.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Unit of product	Inspection level	AQL
Needle	Hardness	3.2.4	GG-I-526	1-needle	S-2	1.5
	Salt Spray*	GG-I-526	GG-I-526	1-needle	S-2	1.5
	Boil test	GG-I-526	GG-I-526	1-needle	S-2	1.5
	Security of assembly of cannula	3.4	4.4.1	1-needle	S-2	1.5
	Diameter & depth	3.5.1	4.4.2.1	1-needle	s-2	1.5
	Freedom from foreign matter	3.5.2	4.4.2.2	1-needle	S-2	1.5
	Fluid tightness	3.5.3	4.4.2.3	1-needle	S-2	1.5

*Test required on brass (plated) parts only.

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4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the needles shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any needles in the sample containing one or more defects shall be rejected and, if the number of defective needles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Style and design not as specified.
102	Hub not Luer-lock.
103	Chuck does not grip cannula.
104	Needles not sharp.
105	Needles not free of feathered edges, burrs or other defects.
Minor	
201	Finish not as specified.
202	Instrument not permanently and legibly marked with the name or trademark of the contractor.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Security of assembly of cannula proper to hub. The assembly of the cannula proper to the hub shall be such that the cannula proper will not pull out of the hub when a gradual pull of 25 pounds is applied for one minute.

4.4.2 Hub cavity.

4.4.2.1 Diameter and depth of hub cavity. The Luer hub cavity shall be tested with the male Luer taper plug gage in accordance with Figure 5 of ANSI Standard Z70.1 (this gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.2.3. The hub shall be placed onto the plug gage with a straight motion, without turning or twisting the plug gage into the cavity.

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4.4.2.1 Diameter and depth of hub cavity. (Cont'd) Once the female Luer cavity is placed onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer hub cavity onto this gage shall be tested by means of the feeler gage in accordance with Figure 5 and 6 of ANSI Standard Z70.1.

4.4.2.2 Freedom from foreign matter. The hub cavity shall be tested for freedom from foreign matter by visual examination under 7x magnification and good illumination; and shall also show complete absence of metal chips and other foreign matter upon visual examination, under a magnifier, of the cotton fibers of a cotton swab (tightly wrapped on the end of small diameter wooden applicator) which has been vigorously turned down into the bottom of the hub cavity.

4.4.2.3 Fluid tightness test. The plug gage specified in 4.4.2.1 shall be inserted into the female Luer hub cavity with a straight motion, without turning or twisting the plug gage into the cavity with a force not to exceed 5 pounds (with the needle in a horizontal position). The feeler gage shall be placed onto the plug gage to assure that the "GO" position is not violated. This arrangement shall then be assembled to a water pressure source having a pressure gage. The water shall be permitted to run through the needle so as to fill the hub cavity space and the lumen of the cannula. The lumen shall be sealed in a leak proof manner. Water, at a pressure of 45 pounds per square inch gage shall be applied for 10 seconds.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials (Packaging and packing)	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. Each biopsy needle, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be thoroughly cleaned and packaged in a box of an appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. The unit shall be protected within the box by being assembled on a substantial cardboard or plastic rack or the box shall contain suitable interior fittings so designed that the needle is held securely in place in such a manner that the point cannot be damaged by contact with the container. The box shall be secured and wrapped in a clear sealed plastic barrier.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636 type CF, class domestic. The box shall be secured to prevent opening under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. However, each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B. Twenty four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Marking may be printed on the wrap, on the box or on a label inserted within the bag in such a manner so as to permit ready identification. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The biopsy needle is intended for use in obtaining a specimen from the kidneys.

6.2 Ordering data. The procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-069-5905	NEEDLE BIOPSY, Kidney, Menghini, 6-1/8 Inch.

It does not include all types as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes requirements of Purchase Description No. 4, dated 18 January 1967 for NSN 6515-00-069-5905 NEEDLE, BIOPSY, Kidney, Menghini, 6-1/8 Inch.

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Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1918

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-N-37175 NEEDLE, BIOPSY, KIDNEY, MENGHINI, 6-1/8 Inch

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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